

## Task 4: Inheritance Based Application

### Aims:-

To design and implement a simple vehicle Rental System using inheritance in Java, where different vehicle types calculate their cost based on a base rent.

### Algorithm:-

1. Start the program.
2. Create a base class vehicle with the following data members
  - Vehicle Id
  - model name
  - Base rent
3. Define a constructor in the vehicle class to initialize the values.
4. Create a method display() in the base class to display vehicle details.
5. Create a derived class car that inherits from vehicle.
6. Create another derived class bike that inherits from vehicle.
7. In the main() method,
  - Create an object of car and display its details and total rent.
8. Stop the program.

### Program:-

```
Class Vehicle {  
    int vehicle id;  
    String Model Name;  
    double base rent;
```

```

vehicle(int vehicle id, String model name, double
        base Rent){
    this.vehicle Id = vehicle Id;
    this.vehicle Id = base Rent;
}
void display() {
    system.out.println("Vehicle ID:" + vehicle ID);
    system.out.println("Model Name:" + Model Name);
    system.out.println("Base Rent:" + base Rents);
}
}

class car extends vehicle {
    car(int vehicle Id, String model Name, double base
        Rent){
        super(vehicle Id, model Name, base Rent);
    }
    double calculate Rent() {
        return base Rent + 200;
    }
}

public class vehicle Rental {
    public static void main(String[] args) {
        Car car = new car(101, "Honda City", 2000);
        car.display();

        System.out.println();
        Bike bike = new bike(201, "Yamaha R15", 200);
        System.out.println("... Car Details ...");
        bike.display();
    }
}

```

### Input :-

car details :-

Vehicle ID : 101

Model Name : Honda City

Base Rent : 2000.0

Bike Details :-

Vehicle ID : 201

Model Name : Yamaha R15

Base Rent : 800

### Output :-

--- Car Details ---

Vehicle ID : 101

Model Name : Honda City

Base Rent : 2000.0

Total Rent : 2500.0

--- Bike Details ---

Vehicle ID : 201

Model Name : Yamaha R15

Base Rent : 800.0

Total Rent : 1000.0

```

System.out.println("Total Rent");
System.out.println("Total Rent:" + bike
    calculateRent());

```

}  
}

| VEL TECH - CSE          |    |
|-------------------------|----|
| EX NO.                  | 4  |
| PERFORMANCE (5)         | 5  |
| RESULT AND ANALYSIS (5) | 5  |
| VIVA VOCE (5)           | 5  |
| RECORD (5)              | 5  |
| TOTAL (20)              | 20 |
| SIGN WITH DATE          |    |

Result:-

This a vehicle Rental system was successfully implemented using inheritance in Java, and the rental cost for car and Bike was calculated correctly.